

6	ai	Total resistance of the 4 parallel resistors = $(1/2R + 1/2R)^{-1} = R$. $I = E/(R + r)$.	B1 A1
	aii	Power in the 4 parallel resistors = $I^2 R$, where $I = E/(R+r)$ Power in the battery = IE $\frac{\text{power in external resistors}}{\text{power by batter}} = \frac{I^2 R}{IE} \text{ or } \frac{I^2 R}{I^2 (R+r)}$ $= \frac{I R}{E}$ $= \frac{E R}{R + r} \frac{1}{E}$ $= \frac{R}{R + r}$	M1 A0
	aiii	Having 4 resistors in a network shown in Fig. 6.1, each resistor only dissipates $\frac{1}{4}$ of the power compared to a single resistor. Hence one of the resistors in Fig. 6.1 is less likely to exceed the power rating.	B1
	bi	There will be no deflection in the galvanometer when the p.d across JY = 1.5 V. Let the balance length (between J and Y) be L . Then $V_{JY} = \frac{L \times 1.50}{(1.50+0.50)} \times 4.0 = 1.5$ $L = 0.500 \text{ m.}$	C1 A1
	bii	When 4.7Ω connected with the 1.5 V cell, the terminal p.d across the 4.7Ω is $V = \frac{4.7}{4.7+0.50} \times 1.5 = 1.356 \text{ V}$ There will null deflection when $V_{JY} = 1.356 \text{ V}$. Then $V_{JY} = \frac{L' \times 1.50}{(1.50+0.50)} \times 4.0 = 1.356$ $L' = 0.452 \text{ m.}$ Alternative working: $L' = \frac{1.356}{3} \times 1.0 = 0.452 \text{ m}$	C1 C1 A1

7	2	[Similarity] The discrete lines of both absorption and emission spectrum	P1
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